

Math Camp Exam Solutions 2021

1.

(a) True. For any $S \subset X$, $x \in X$ is a limit point if $\forall \varepsilon > 0$,

$$(B_\varepsilon(x) \setminus \{x\}) \cap S \neq \emptyset.$$

Suppose S has finite number of elements and x is a limit point of S . Let $S := \{x_1, \dots, x_n\}$. WLOG, assume $x \neq x_i$ for any $i \in \{1, 2, \dots, n\}$. Let $r = \min\{d(x, x_1), d(x, x_2), \dots, d(x, x_n)\}$. Then,

$$(B_r(x) \setminus \{x\}) \cap S = \emptyset \text{ (contradiction!)}$$

If $x \in \{x_1, \dots, x_n\}$, carry out the same exercise as above with

$$r = \min\{d(x, x_j) : x_j \in S \setminus \{x\}\}.$$

Other steps remain the same.

(b) False. A zero matrix is diagonal (and hence diagonalizable) but clearly not invertible.

(c) True. Let $(x_n)_{n \geq 1}$ be a convergent sequence. To show $(x_n)_{n \geq 1}$ is Cauchy, we need to show $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall m, n \geq N, d(x_m, x_n) < \varepsilon$.

Fix $\varepsilon > 0$. Since $(x_n)_{n \geq 1}$ is convergent,

$$\implies \exists N \in \mathbb{N} \text{ s.t. } \forall k \geq N, d(x_k, x) < \varepsilon/2$$

$$\implies \forall m, n \geq N, d(x_m, x_n) \leq d(x_m, x) + d(x_n, x) < \varepsilon.$$

Thus, $(x_n)_{n \geq 1}$ is Cauchy.

(d) False. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^2$. Let $E = (-1, 1)$. Then $f(E) = [0, 1)$ is not open although f is continuous (under standard metric).

(e) False. Consider $(\mathbb{R}_{++}, d_2)$ and let $S = (0, 1]$. Clearly S is closed and bounded in $\mathbb{R}_{++}$.

Consider a family of open sets $\{E_n\}_{n \geq 1}$ where $E_n = (\frac{1}{n}, 2)$. Clearly,

$$\bigcup_{n \geq 1} E_n \supset (0, 1],$$

i.e., $\{E_n\}_{n \geq 1}$ is an open cover of $(0, 1]$. However, there does not exist any finite subcover because for each $n \geq 1$, there is $a \in \mathbb{R}$ such that $0 < a < \frac{1}{n}$.

(f) True. $T : X \rightarrow X$ is a contraction if $\exists \lambda < 1$ such that

$$d(T(x), T(y)) \leq \lambda d(x, y), \quad \forall x, y \in X.$$

For $T_1 \circ T_2 : X \rightarrow X$, $(T_1 \circ T_2)(x) = T_1(T_2(x))$,

$$d(T_1 \circ T_2(x), T_1 \circ T_2(y)) \leq \lambda d(T_2(x), T_2(y)) \leq \lambda d(x, y), \quad \forall x, y \in X.$$

Thus, $T_1 \circ T_2$ is a contraction.

Similarly for $T_2 \circ T_1 : X \rightarrow X$, $(T_2 \circ T_1)(x) = T_2(T_1(x))$,

$$d(T_2 \circ T_1(x), T_2 \circ T_1(y)) \leq d(T_1(x), T_1(y)) \leq \lambda d(x, y), \quad \forall x, y \in X.$$

Therefore, $T_2 \circ T_1$ is a contraction.

(g) True. Let $\{E_\alpha\}_{\alpha \in A}$ be a family of compact sets. Then,

$$\begin{aligned} & \forall \alpha \in A, E_\alpha \text{ is compact} \\ \implies & \forall \alpha \in A, E_\alpha \text{ is closed} \\ \implies & \bigcap_{\alpha \in A} E_\alpha \text{ is closed.} \end{aligned}$$

Since $\bigcap_{\alpha \in A} E_\alpha \subset E_{\alpha_0}$ for some $\alpha_0 \in A$,

$$\begin{aligned} & \bigcap_{\alpha \in A} E_\alpha \text{ is a closed subset of a compact set} \\ \implies & \bigcap_{\alpha \in A} E_\alpha \text{ is a compact set.} \end{aligned}$$

(h) False. Consider $(x_n)_{n \geq 1}$ such that $x_n = \frac{1}{n}$ and $(y_n)_{n \geq 1}$ such that $y_n = -\frac{1}{n}$. Clearly $x_n > y_n, \forall n \in \mathbb{N}$, although $x_n \rightarrow 0$ and $y_n \rightarrow 0$. Thus, $x = y$ but not $x > y$.

2.

By the formula for characteristic polynomial,

$$\begin{aligned}p(\lambda) &= \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) \\&= \lambda^2 - 7\lambda - 8 \\&= (\lambda - 8)(\lambda + 1) = 0.\end{aligned}$$

Therefore $\lambda_1 = -1$ and $\lambda_2 = 8$, and

$$\Lambda = \begin{bmatrix} -1 & 0 \\ 0 & 8 \end{bmatrix}.$$

P will comprise the corresponding eigenvectors. For $\lambda_1 = -1$,

$$\begin{aligned}(A + I)x &= 0 \\ \Rightarrow \begin{bmatrix} 6 & -3 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow x_1 &= \frac{1}{2}x_2 \\ \Rightarrow \text{eigenvector corresponding to } \lambda_1 = -1 &: \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \text{ (or any vector } \beta \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \beta \in \mathbb{R} \setminus \{0\}.)\end{aligned}$$

Likewise, eigenvector corresponding to $\lambda_2 = 8$ is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Thus,

$$P = \begin{bmatrix} 1/2 & 1 \\ 1 & -1 \end{bmatrix},$$

and A is diagonalizable in $\mathbb{R}$.

$$A^n = P\Lambda^n P^{-1} = \begin{bmatrix} 1/2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} (-1)^n & 0 \\ 0 & 8^n \end{bmatrix} \begin{bmatrix} 1/2 & 1 \\ 1 & -1 \end{bmatrix}^{-1}.$$

A^n does not converge to some real matrix B as $n \rightarrow \infty$ since $|\lambda_2| = 8 > 1$. A is neither positive nor negative definite because $\lambda_1 < 0$ and $\lambda_2 > 0$.

3.

(a)

$$\begin{aligned}
 F(\lambda X_1, \lambda X_2, \dots, \lambda X_n) &= \sum_{i=1}^n \alpha_i (\lambda X_i)^\gamma \\
 &= \alpha_1 X_1^\gamma \lambda^\gamma + \alpha_2 X_2^\gamma \lambda^\gamma + \dots + \alpha_n X_n^\gamma \lambda^\gamma \\
 &= \lambda^\gamma (\alpha_1 X_1^\gamma + \alpha_2 X_2^\gamma + \dots + \alpha_n X_n^\gamma) \\
 &= \lambda^\gamma \sum_{i=1}^n \alpha_i X_i^\gamma \\
 &= \lambda^\gamma F(X_1, \dots, X_n).
 \end{aligned}$$

F is homogeneous of degree γ and therefore homothetic.

(b)

$$\frac{\partial}{\partial x_i}(x_i^\gamma) = \gamma x_i^{\gamma-1}, \quad \frac{\partial^2}{\partial x_i^2}(x_i^\gamma) = \gamma(\gamma-1)x_i^{\gamma-2}$$

When $\gamma < 0$, then $\frac{\partial^2}{\partial x_i^2}(x_i^\gamma) > 0$ because $\gamma(\gamma-1) > 0$. Thus, x_i^γ is convex and so is $\sum_{i=1}^n \alpha_i x_i^\gamma$.

When $\gamma \in (0, 1)$, then

$$\frac{\partial^2}{\partial x_i^2}(x_i^\gamma) < 0 \implies \text{function is concave. (the same logic as before)}$$

When $\gamma > 1$,

$$\frac{\partial^2}{\partial x_i^2}(x_i^\gamma) > 0 \implies \text{function is convex.}$$

At $\gamma = 1$, the function is linear and therefore both concave and convex.

(c)

$$\begin{aligned}
 Y &= \sum_{i=1}^n \alpha_i X_i^\gamma, \\
 \implies \ln Y &= \ln \left(\sum_{i=1}^n \alpha_i X_i^\gamma \right) \\
 \implies \ln Y^* &= \ln \left(\sum_{i=1}^n \alpha_i (X_i^*)^\gamma \right) \quad \text{and} \quad Y^* = \sum_{i=1}^n \alpha_i (X_i^*)^\gamma.
 \end{aligned} \tag{1}$$

Taking Taylor approximations on both sides of (1),

$$\begin{aligned}
\ln Y^* + \frac{Y - Y^*}{Y^*} &= \ln \left(\sum_{i=1}^n \alpha_i (X_i^*)^\gamma \right) + \frac{1}{\sum_{i=1}^n \alpha_i (X_i^*)^\gamma} \left[\sum_{i=1}^n \alpha_i \gamma (X_i^*)^{\gamma-1} (X_i - X_i^*) \right] \\
\Rightarrow \ln Y^* + \frac{Y - Y^*}{Y^*} &= \ln Y^* + \frac{1}{Y^*} \left[\sum_{i=1}^n \alpha_i \gamma (X_i^*)^{\gamma-1} (X_i - X_i^*) \right] \\
\Rightarrow \frac{Y - Y^*}{Y^*} &= \frac{1}{Y^*} \left[\sum_{i=1}^n \alpha_i \gamma (X_i^*)^\gamma \frac{X_i - X_i^*}{X_i^*} \right] \\
\Rightarrow \hat{Y} &= \frac{1}{Y^*} \left[\sum_{i=1}^n \alpha_i \gamma (X_i^*)^\gamma \hat{X}_i \right] \quad \text{or} \quad \hat{Y} = \sum_{i=1}^n \frac{\alpha_i \gamma (X_i^*)^\gamma}{\sum_{i=1}^n \alpha_i (X_i^*)^\gamma} \hat{X}_i.
\end{aligned}$$

4.

- (a) Not upper hemi-continuous at x_1 and x_2 .
- (b) Lower hemi-continuous at both x_1 and x_2 .
- (c) Yes. Yes. Yes.
- (d) No.

5.

- (a) Yes. The utility function $u(x_1, x_2) := x_1^{1/2} + x_2^{1/2}$ is continuous, and

$$D := \{(x_1, x_2) : p_1 x_1 + p_2 x_2 \leq m, x_1 \geq 0, x_2 \geq 0\}$$

is bounded by $B_\varepsilon(0)$ where $\varepsilon = \max\{m/p_1, m/p_2\} + 1$ and closed since if $(x_1^n, x_2^n) \rightarrow (x_1, x_2)$ and $(x_1^n, x_2^n) \in D$, then $(x_1, x_2) \in D$. (Easy to show by the properties of sequences.

$$\begin{aligned}
(x_1^n, x_2^n) \in D, \forall n &\implies x_1^n \geq 0, x_2^n \geq 0, \forall n, \quad \text{and} \\
(x_1^n, x_2^n) \rightarrow (x_1, x_2) &\implies x_1^n \rightarrow x, x_2^n \rightarrow x_2 \\
\therefore x_1^n \geq 0, \forall n &\implies x_1 \geq 0, x_2^n \geq 0, \forall n \implies x_2 \geq 0.
\end{aligned}$$

Similarly,

$$\begin{aligned}
x_1^n \rightarrow x, x_2^n \rightarrow x_2 &\implies p_1 x_1^n + p_2 x_2^n \rightarrow p_1 x_1 + p_2 x_2 \\
\therefore p_1 x_1^n + p_2 x_2^n \leq m, \forall n &\implies p_1 x_1 + p_2 x_2 \leq m.
\end{aligned}$$

By Heine-Borel theorem, D is compact. By Weierstrass theorem, a maximum exists, and therefore a solution exists.

(b)

$$\begin{aligned}\frac{\partial}{\partial x_1}u(x_1, x_2) &= \frac{1}{2}x_1^{-1/2} > 0, \quad \forall x_1 \geq 0, \\ \frac{\partial}{\partial x_2}u(x_1, x_2) &= \frac{1}{2}x_2^{-1/2} > 0, \quad \forall x_2 \geq 0,\end{aligned}$$

and hence $u(x_1, x_2)$ is monotonic. Therefore, the budget constraint must bind.

Alternatively, let (x_1^*, x_2^*) be in argmax such that $p_1x_1^* + p_2x_2^* < m$. Consider

$$(\hat{x}_1, \hat{x}_2) = (x_1^* + (m - p_2x_2^*)/p_1, x_2^*).$$

By construction,

$$\hat{x}_1 > x_1^* \implies u(\hat{x}_1, \hat{x}_2) > u(x_1^*, x_2^*),$$

so (x_1^*, x_2^*) was not an optimum.

(c)

$$\mathcal{L} = x_1^{1/2} + x_2^{1/2} + \lambda_1(m - p_1x_1 - p_2x_2) + \lambda_2x_2 + \lambda_3x_3.$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{1}{2}x_1^{-1/2} - \lambda_1p_1 + \lambda_2 = 0, \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{1}{2}x_2^{-1/2} - \lambda_1p_2 + \lambda_3 = 0. \quad (3)$$

CSC: (complementary slackness condition)

$$\lambda_1(m - p_1x_1 - p_2x_2) = 0 \quad \text{with } \lambda_1 \geq 0 \text{ and } m - p_1x_1 - p_2x_2 \geq 0,$$

$$\lambda_2x_1 = 0 \quad \text{with } \lambda_2 \geq 0 \text{ and } x_1 \geq 0,$$

$$\lambda_3x_2 = 0 \quad \text{with } \lambda_3 \geq 0 \text{ and } x_2 \geq 0.$$

Soln 1: At optimum, $m - p_1x_1 - p_2x_2 = 0$.

Case 1: $\lambda_1 \geq 0$, $\lambda_2 > 0$ and $\lambda_3 > 0$. By CSC, $x_1 = 0$ and $x_2 = 0$. This violates $m - p_1x_1 - p_2x_2 = 0$.

Case 2: $\lambda_1 \geq 0$, $\lambda_2 \geq 0$ and $\lambda_3 > 0$. By CSC, $x_2 = 0$. Using (3), we get

$$\infty - \lambda_1p_2 + \lambda_3 = 0,$$

which is not possible.

Case 3: $\lambda_1 \geq 0$, $\lambda_2 > 0$, $\lambda_3 \geq 0$. By CSC, $x_1 = 0$. By symmetry, this case is also not possible.

Case 4: $\lambda_1 \geq 0, \lambda_2 = \lambda_3 = 0$. By (2) and (3), we have

$$\begin{aligned}
\frac{1}{2}x_1^{-1/2} - \lambda_1 p_1 &= 0, \\
\frac{1}{2}x_2^{-1/2} - \lambda_1 p_2 &= 0, \\
\Rightarrow \frac{1}{2}x_1^{-1/2} \frac{1}{p_1} &= \frac{1}{2}x_2^{-1/2} \frac{1}{p_2} \\
\Rightarrow \left(\frac{x_1}{x_2}\right)^{-1/2} &= \frac{p_1}{p_2} \\
\Rightarrow x_2 &= \left(\frac{p_1}{p_2}\right)^2 x_1.
\end{aligned}$$

$$\begin{aligned}
m - p_1 x_1 - p_2 x_2 &= 0 \\
\Rightarrow p_1 x_1 + p_2 \left(\frac{p_1}{p_2}\right)^2 x_1 &= m \\
\Rightarrow x_1 \left(p_1 + \frac{p_1^2}{p_2}\right) &= m \\
\Rightarrow x_1 &= \frac{mp_2}{p_1(p_1 + p_2)} \\
\Rightarrow x_2 &= \left(\frac{p_1}{p_2}\right)^2 x_1 = \left(\frac{p_1}{p_2}\right)^2 \frac{mp_2}{p_1(p_1 + p_2)} = \frac{mp_1}{p_2(p_1 + p_2)}.
\end{aligned}$$

$$\begin{aligned}
\lambda_1 &= \frac{1}{2}x_1^{-1/2} \frac{1}{p_1} = \frac{1}{2p_1} \left(\frac{mp_2}{p_1(p_1 + p_2)}\right)^{-1/2} \\
\Rightarrow \lambda_1 &= \frac{1}{2} \frac{m^{-1/2}}{p_1^{1/2} p_2^{1/2}} (p_1 + p_2)^{1/2} \\
&= \frac{1}{2} m^{-1/2} \left(\frac{p_1 + p_2}{p_1 p_2}\right)^{1/2} \\
&= \frac{1}{2} m^{-1/2} \left(\frac{1}{p_1} + \frac{1}{p_2}\right)^{1/2}.
\end{aligned}$$

$$\therefore (x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(\frac{mp_2}{p_1(p_1 + p_2)}, \frac{mp_1}{p_2(p_1 + p_2)}, \frac{1}{2} m^{-1/2} \left(\frac{1}{p_1} + \frac{1}{p_2}\right)^{1/2}, 0, 0\right).$$

Soln 2: Argue that at optimum $x_1^* \neq 0$ and $x_2^* \neq 0$ by looking at the partial derivatives

$$\frac{\partial}{\partial x_i} (x_1^{1/2} + x_2^{1/2}) = \frac{1}{2} x_i^{-1/2} \Big|_{x_i=0} = \infty \text{ (marginal utility is } \infty \text{)}.$$

Thus, Case 1, 2 and 3 are irrelevant.

(d) Since $p_1, p_2 > 0$, the only way CQ fails is if all three constraints simultaneously bind, which is not possible from part (c). This is because

$$\begin{aligned}\nabla(m - p_1x_1 - p_2x_2) &= (-p_1, -p_2), \\ \nabla(x_1) &= (1, 0), \\ \nabla(x_2) &= (0, 1).\end{aligned}$$

(e) From Case 4, the indirect utility function is

$$\begin{aligned}v(p_1, p_2, m) &= u(x_1^*, x_2^*) \\ &= \left(\frac{mp_2}{p_1(p_1 + p_2)} \right)^{1/2} + \left(\frac{mp_1}{p_2(p_1 + p_2)} \right)^{1/2} \\ &= \left(\frac{m}{p_1 + p_2} \right)^{1/2} \left[\left(\frac{p_2}{p_1} \right)^{1/2} + \left(\frac{p_1}{p_2} \right)^{1/2} \right] \\ &= \left(\frac{m}{p_1 + p_2} \right)^{1/2} \left[\frac{p_2 + p_1}{p_1^{1/2} p_2^{1/2}} \right] \\ &= \left(\frac{m^{1/2}}{p_1^{1/2} p_2^{1/2}} \right) (p_1 + p_2)^{1/2}.\end{aligned}$$

$$\therefore \frac{\partial v}{\partial m}(p_1, p_2, m) = \frac{1}{2} m^{-1/2} \left(\frac{p_1 + p_2}{p_1 p_2} \right)^{1/2} = \frac{1}{2} m^{-1/2} \left(\frac{1}{p_1} + \frac{1}{p_2} \right)^{1/2}.$$

By differentiating the Lagrangian function,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial m}(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) &= \lambda_1^* \\ &= \frac{1}{2} m^{-1/2} \left(\frac{1}{p_1} + \frac{1}{p_2} \right)^{1/2},\end{aligned}$$

which coincides with $\frac{\partial v}{\partial m}(p_1, p_2, m)$. This confirms the envelope theorem.